

Marco Antonio de Souza Carvalho

Computer Engineering Student — Backend, APIs and AI Systems

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PROFESSIONAL SUMMARY

I build practical projects with Python, JavaScript/Node.js, REST APIs and service integrations. I am currently directing my training toward Java backend development and Spring Boot, with a focus on quality, security and continuous improvement.

I am seeking an internship or junior role in backend, data engineering, AI or cloud. I combine Computer Engineering fundamentals with projects that are published, documented and version-controlled in an open repository.

EDUCATION

Computer Engineering

Centro Universitário de Brasília (UniCeub) · Jul 2024 – expected 2028 · currently in the 4th semester

Completed coursework: Introduction to Computing, Programming Logic, Computer Organization and Architecture, Software Engineering, Databases I and II, Calculus for Computing, Linear Algebra and Analytic Geometry, Digital Circuits, Computer Networks, Cybersecurity, Communication Systems and Computer-Mediated Design.

Academic exchange

University of Aveiro, Portugal · Jul 2022 – Sep 2023

International academic experience, using English daily for study and collaboration.

TECHNICAL SKILLS

Development and data

Python (intermediate) · JavaScript / Node.js · Java and Object-Oriented Programming · Express and REST APIs · FastAPI and LLM APIs · SQL and data modeling · Software Engineering · Git / GitHub · C++ (basic to intermediate)

Networking, security and hardware

Computer networks and security · Cisco Cybersecurity · Linux fundamentals · Basic electronics · Soldering and prototyping · Multimeter and schematic reading

Currently learning

AWS (IAM, EC2, S3, VPC, CloudWatch, Lambda) · Spring Boot · Apache Kafka · Microservices

Items marked "currently learning" are active study paths, not certifications or completed professional experience.

PROJECTS

AI Investment System — Python, FastAPI

Multi-agent system in development that orchestrates specialized agents (technical, fundamental, sentiment and macroeconomic) to produce auditable investment theses. Includes portfolio optimization (Markowitz and Risk Parity), risk controls and alerts through the Telegram Bot API.

AI Job Apply — Node.js, OAuth, GitHub Actions

Pipeline that matches job requirements against a candidate profile without acting on its own: OAuth authentication with state validation, revocable consent through a dedicated endpoint, and mandatory human approval before any submission. Continuous integration runs in *dry-run* mode only, with a scheduled smoke test monitoring the public demo.

Ticket Booking System — Java

Ticket reservation application covering domain modeling, inheritance, polymorphism, exception handling and file persistence.

Personal portfolio — HTML, CSS, JavaScript

Bilingual, responsive site published through GitHub Pages, with automated HTML validation and post-deploy verification in continuous integration.

HANDS-ON LAB EXPERIENCE

Hands-on practice from Computer Engineering coursework — this does not correspond to professional work experience.

- Electronic board prototyping: manual component assembly and functional testing.
- Soldering and low-voltage circuit assembly.
- Documentation of procedures and results with Git/GitHub.
- Reading and translating technical manuals in English.

LANGUAGES

Portuguese: native. **English:** fluent in reading, writing and conversation (one year abroad and Brasas language school). **Spanish:** basic comprehension.

ADDITIONAL INFORMATION

Available Monday to Friday · Immediate start · Driver's license (category B).